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Aspect-based sentiment classification of user reviews to understand customer satisfaction of e-commerce platforms

Laleh Davoodi[1] · **József Mezei**[1] · **Markku Heikkilä**[1]

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Abstract

Making use of user-generated content, in particular customer reviews, is an essential means for companies to gather information about customers. In order to utilize reviews in improving customer satisfaction, companies need to identify the important components of their business that are discussed by the customers. In this article, we propose to use aspect-based sentiment analysis for reviews focusing on e-commerce platforms. Fourteen aspects were extracted from the literature and empirical analysis of 3500 randomly selected reviews from the Trustpilot platform. A unique dataset was created by manually annotating the reviews and assigning a sentiment for each aspect-review pair. Aspect-based sentiment classification was performed using some of the most recent machine learning models, with RoBERTa identified as the best-performing model, achieving over 90% accuracy. We also demonstrate how insights generated from the sentiment scores can assist companies in improving their service and increasing customer satisfaction.

Keywords Sentiment analysis · Machine learning · Deep learning · Customer satisfaction · RoBERTa

✉ József Mezei
jozsef.mezei@abo.fi

Laleh Davoodi
laleh.davoodi@abo.fi

Markku Heikkilä
markku.heikkila@abo.fi

[1] Faculty of Social Sciences, Business and Economics, and Law, Åbo Akademi University, Turku, Finland

1 Introduction

The exponential growth of the internet has led to information overload, particularly in the context of e-commerce [1]. It has become common practice for various online platforms to offer customers the opportunity to share feedback, which is then published on the website. These reviews are useful for prospective customers when evaluating their purchasing decisions [2] (when choosing a product or even a platform to purchase from) based on recommendations [3]. These reviews can also be an important source of information for companies seeking to understand the factors driving customer satisfaction and identifying the most significant pain points that may trigger customer churn. Furthermore, important qualitative aspects of online user-generated content, specifically customer reviews, reflect customers' attitudes toward brands [4].

The study of consumer reviews not only benefits businesses by informing product development and marketing strategies but also enriches academic understanding of consumer behaviour in the e-commerce sector and contributes to the advancement of textual data analysis techniques. Numerous studies [5] have analysed the relationship between customer satisfaction and various characteristics of the purchasing process on e-commerce platforms. Research by [6] found that the quality of logistics services and price fairness are closely related to customer satisfaction and repurchase intention. Griva [7] presents the analysis of a large-scale customer survey using an unsupervised machine learning algorithm, focusing on the main components of customer satisfaction. The article identifies different clusters of customers, each attributing varying levels of importance to components such as Product Availability, Prices, and Usability.

Natural Language Processing (NLP) tools are increasingly used to understand customer satisfaction. For example, [8] illustrates how deep learning models can identify the most important drivers of e-commerce satisfaction from open-ended surveys. However, there is limited research on the insights that can be gained from online customer reviews analysed using advanced natural language processing tools aimed at discovering not only the drivers but also the associated sentiment of customers [9]. The limited availability of research can be attributed to multiple issues. First, the inherent complexity of advanced NLP techniques, particularly those involving a deep understanding of machine learning and transformer models, poses a significant barrier. Second, accessing high-quality, large-scale datasets of customer reviews is challenging. Such datasets are essential for training and testing NLP models, however, issues related to data collection, preprocessing, and the need for extensive data cleaning complicate their acquisition and use. This also involves the problem of acquiring high-quality annotation of datasets with carefully selected aspects of the e-commerce domain. Third, the computational resources required for running these advanced models are substantial. Additionally, the rapid evolution of NLP methodologies means that keeping pace with the latest advancements is a constant challenge, often requiring continuous learning and adaptation. These factors collectively contribute to the limited research in this area, despite the growing recognition of its importance and potential.

To address this problem, the main research goals of this study are: (i) to identify the most important dimensions/aspects of the purchase process on an e-commerce platform, (ii) to prepare a dataset of customer reviews annotated according to the identified aspects, (iii) to identify the best-performing machine learning models for delivering reliable sentiment predictions in relation to the aspects, and (iv) to generate actionable business insights connecting aspects and customer satisfaction. This article presents a systematic study that combines these four goals using deep-learning models for aspect-based sentiment analysis in e-commerce. The study utilizes a manually annotated dataset of customer reviews with carefully selected aspects covering the key drivers of customer satisfaction.

In this study, to address the proposed objectives, we focus on performing aspect-based sentiment analysis on online reviews written by e-commerce platform customers. We collected customer reviews from the online platform Trustpilot, focusing on five e-commerce platforms (Zalando, Wish, Sheinside, Boozt, and Nelly). These platforms were selected for their substantial online presence; at the time of writing, none of them operated a physical store. They all focus on a wide range of fashion and lifestyle products. Additionally, these platforms have an international user base. The reviews collected from these e-commerce platforms offer a comprehensive view of the components of customer satisfaction across different market segments and geographical regions.

Based on empirical analysis of the dataset and a detailed literature review, we identified 14 aspects that cover all the important stages of the e-commerce purchase process and the 'pain points' that determine customer satisfaction. Most articles developing sentiment analysis models use computational methods and algorithms to assign a sentiment value to larger datasetsm typically employing: (i) statistical models based on the frequency of a predefined set of sentiment words, or (ii) rule-based or more complex machine learning algorithms. In contrast, we manually annotated approximately 3000 reviews with respect to all 14 aspects. To the best of our knowledge, this is the first dataset manually annotated at this level of detail in the e-commerce domain. The descriptive results and the performance of state-of-the-art deep learning aspect-based sentiment classification models demonstrate that we can generate useful and reliable insights. These insights provide a detailed understanding of the different stages of the purchase process, helping companies improve their offering and retain customers.

The rest of the paper is structured as follows. In Sect. 2, we summarize related research on customer satisfaction, e-commerce, and (aspect-based) sentiment analysis. Section 3 presents the research methodology and discusses data collection and preparation. The results of sentiment classification and a descriptive analysis of aspects from the perspective of customer satisfaction are presented in Sect. 4. Finally, Sect. 5 provides the conclusions, limitations, and directions for future research.

2 Literature review

In this section, we summarize the academic literature relevant to various components of the presented research. We start with e-commerce and customer satisfaction, focusing on identifying the most important determinants of customer sentiment

as they are used in online review analysis. Related literature on (aspect-based) sentiment analysis, particularly as performed using deep learning models, is also discussed.

2.1 E-commerce

The rise of e-commerce, alongside increased digitalization and competition, has provided businesses withan unprecedented opportunity to capitalize on business-to-consumer (B2C) interactions by enabling them to create, manage, and attempt to control this relationship with each customer. E-commerce is highly profitable but equally competitive, and as shown for example by [10], the key to staying competitive and retaining existing customers is high-quality customer service. This crucial role stems from the observation that in e-commerce, there is a high likelihood of post-purchase dissonance since customers cannot experience the products in person. For example, as [11] points out, customer service is positively related to customer satisfaction in an e-commerce setting. Similarly, [12] found that improved customer service is a key driver of the recent increase in e-commerce and online shopping. In the current digital age, as many companies adopt automated, AI-based customer service solutions, such as chatbots [13], ensuring customer satisfaction becomes even more crucial.

As a customer activity in e-commerce platforms, e-shopping refers to the B2C part of e-commerce, where customers purchase or obtain knowledge about consumer goods online. The decision to continue or abandon the established relationship with an e-commerce platform depends on several factors related to specific stages of the customer journey [14]. Generally, the process can be divided into the pre-purchase, purchase, and post-purchase stages, within which numerous touch-points have an impact customer experience and satisfaction. According to [14], touch-points can further be analysed from the perspective of which stakeholders in the process 'own' the related decisions.

Considering the products purchased on e-commerce platforms, the most frequently mentioned factors influencing customer satisfaction include pricing and product and brand availability. Price is one of the most powerful elements in e-commerce, directly impacting profitability and customers' purchase decisions [15]. Consumers' perceptions of pricing are reflected in what they are willing to pay for a product and how they perceive the e-commerce platform overall; this can be influenced by price promotions and discounts. Furthermore, the presence of popular brands may not only increase customers' willingness to use a platform but also command a higher price premium. According to [16], all aspects of brand leadership positively influence customer satisfaction.

Another important component is the shipping and delivery of products ordered on the e-commerce platform, typically carried out by a third-party logistics company [17]. As [11] notes, the development of advanced shipping and payment solutions is a strong driving force behind the widespread use of e-commerce. Jain et al. [12] identifies several factors related to product shipping that impact customer satisfaction, including timeliness and delivery speed (particularly last-mile delivery),

product condition (including the packaging), and ease of return. However, there is limited research connecting post-purchase activities to customer satisfaction.

2.2 Customer satisfaction

In the current digital age, high-quality customer service is essential for minimizing customer churn and cultivating loyal customers. Customer dissatisfaction, particularly negative online feedback, can significantly harm any business. To systematically address this issue, companies need to focus on customer satisfaction by measuring customer perceptions of their products, services, and qualifications [18].

According to [19], customer satisfaction can be described in two forms: (i) when customers make an emotional decision based on their most recent interaction with a retailer regarding a product or service, or (ii) as long-term cumulative satisfaction derived from the overall customer shopping experience with a certain online retailer. However, according to [20], customer satisfaction is a comparative concept that is always evaluated against a criterion, and accordingly, several competing approaches have been proposed to define it. These theories include the Expectancy-Disconfirmation Paradigm (EDP), Value-Precept Theory, Attribution Theory, Equity Theory, Comparison Level Theory, Evaluation Congruity Theory, Person-Situation-Fit Model, Performance Importance Model, or the Dissonance and the Contrast Theory.

To investigate the origins of customer satisfaction and develop measurements for the construct, a variety of theoretical models have been proposed. The disconfirmation paradigm was employed in the majority of these investigations [21], meaning that satisfaction is shaped by the scope and direction of the disconfirmation experience, which is connected to the individual's preliminary expectations. Individual expectations are established when a product is perceived as desirable, negatively disconfirmed when a product performs worse than anticipated and positively disconfirmed when it exceeds expectations. When a subject's expectations are negatively confirmed, dissatisfaction occurs. Disconfirmation emerges when initial expectations and real performance do not match.

Prior studies on customer satisfaction mainly focused on measuring customer satisfaction itself or the effect of various elements on it. Researchers commonly measure customer satisfaction using methods such as calculating average satisfaction scores, employing advanced statistical analysis, and applying business analytics (BA). Data used in customer satisfaction studies are typically collected from customer ratings, customer surveys, and textual formats, such as customer reviews [7].

2.3 Sentiment analysis

Sentiment Analysis (SA), also known as opinion mining, is a field within Natural Language Processing (NLP) focused on deriving opinions and emotions from unstructured text. Analyzing emotions expressed in online reviews is important for various reasons, including attracting potential customers by guiding their purchasing decisions, receiving product feedback, and even predicting the results of events such as political elections [22]. SA is beneficial for automatically evaluating people's

opinions concerning a product, organization, or service [23]. SA can be challenging due to the complexity of language syntax and semantics, as well as the need to accurately identify and assess sentiment while accounting for the emotions involved. Effective SA requires advanced linguistic analysis techniques and significant efforts in feature extraction for machine learning or rule-based approaches [24]. Since the introduction of SA in academic research, a wide range of studies has been conducted and reported, highlighting its usefulness in generating business insights from emotions [25]. From the perspective of granularity, SA is typically performed at three levels [26]: (i) Document level, (ii) Sentence level, and (iii) Entity/Feature/Aspect level.

In document-level SA, the overall sentiment of the document, considered as a unit, is extracted [27]. Sentence-level SA, on the other hand, focuses on the sentiment of each sentence, treating it as a single entity and predicting the opinion expressed within [28]. In Aspect-Based Sentiment Analysis (ABSA), the task is to extract opinion targets (aspects) and the corresponding sentiment for each target to gain more fine-grained insights [29]. For instance, "*You will like the offering of the e-commerce site A if delivery speed is not the most important thing for you*" is a sentence with a mixed overall sentiment but a negative sentiment regarding the aspect of delivery.

SA techniques are typically divided into two categories: (i) lexicon-based and (ii) machine-learning approaches. The lexicon-based approach involves evaluating the sentiment polarities of a document by analyzing the semantic polarity of the words or terms within it. In the lexicon-based method, a dictionary containing sentiment polarity values for words and phrases is used. The overall sentiment of the text is then calculated based on the majority of these polarities. Dictionaries are typically created manually or automatically by using seed words to extend the list [30]. Moreover, lexicon-based dictionaries may include the emotional classifications for words, such as anger, joy, or fear [31].

In the following, we describe in more detail the main approach utilized in this paper: machine learning-based SA, with a particular focus on deep learning models that gained wide attention in recent years.

2.4 Traditional machine learning and deep learning models for sentiment analysis

Traditional machine learning methods, such as Support Vector Machines (SVM) and Naïve Bayes (NB), are widely applied in various sentiment analysis applications. For example, [32] applied Doc2vec and various machine learning algorithms, such as SVM, Logistic Regression (LR), and NB on 11600 hotel reviews. They found that the best performance (80% precision and 88% recall) was achieved by combining Doc2Vec with SVMs.

In SA, where the data is textual, transforming it into numeric form and extracting key features is a crucial step. Traditional machine learning methods typically use the Bag-of-Words (BoW) or Term Frequency-Inverse Document Frequency (TF-IDF) approach for feature extraction. These methods rely on one-hot word representation

techniques. One disadvantages of these methods is that they create sparse, high-dimensional feature vectors, which lead to significant scalability challenges. More importantly, TF-IDF and BoW models cannot capture word order or the syntactical, and semantic attributes of a sentence [33].

Word-embedding models have been proposed to address the limitations of TF-IDF and BoW feature extraction techniques by extracting both semantic and syntactic elements when representing a word. Word embeddings are typically represented as real-valued vectors in multidimensional space. Word2Vec [34] and GloVe [35] are two widely used embedding methods in which words are represented as real-valued vectors; however, they also come with several limitations despite their high performance. For instance, these methods require a massive corpus for training and producing word embeddings. Moreover, these methods cannot handle out-of-vocabulary words [36]. Another limitation of these models is the assumption that each word maps to exactly one vector representing its meaning [37].

To overcome these limitations and accurately represent context-dependent words, contextualized word embeddings have been proposed. ELMO [38], Flair Embeddings [39], and Bidirectional Encoder Representations from Transformers (BERT) [40] are three recently introduced techniques classified as dynamic word embeddings. These models are known as Pre-trained Language Models (PLM) as they are initially trained on a large unannotated corpus and then fine-tuned on a downstream task. BERT evaluates a word's context simultaneously from both the left and right sides [40]. Thus, it can provide more contextual features from a sentence compared to other models such as ELMO. BERT's bidirectional feature is achieved by masked language modeling (MLM). MLMs mask a random word by replacing it with a token [MASK] in a sequence of input or sentences. The model then attempts to predict the masked words by using the context from both the left and right of the masked word to complete the sentence. In addition to MLM, BERT is also able to predict the next sentence [41].

While ELMO uses a type of Recurrent Neural Network (RNN) called Long Short-Term Memory (LSTM) [42] for word embedding, BERT utilizes transformers [43]. Attention-based mechanisms are a novel architecture used to solve sequence-to-sequence tasks, considering the context-dependency of words at each computational step.

2.5 Aspect-based sentiment analysis

Aspect-based sentiment analysis (ABSA) consists of two sub-tasks: aspect extraction (AE) and aspect sentiment classification (ASC). In our context, performing ABSA provides additional insights into the quality of a given product or service by assessing aggregated sentiments of aspects on websites and e-commerce platforms such as Amazon, Yelp, and Taobao [28]. These reviews typically focus on a single product or service, detailing various features or aspects. ABSA gained special attention as a research method during the 'International Workshop on Semantic Evaluation'(SemEval) in 2014 [44]. Since then, there has been a proliferation of benchmark datasets, notable among them being the dataset by [45], as well as

datasets from the SemEval from 2014 to 2016, covering evaluations for laptops, cameras, and hospitality services [28, 44, 46, 47].

Since then, several articles have been published on theoretical developments and applications of ABSA, both in combination with traditional machine learning models [48–50] and deep learning [36, 51, 52].

Traditional machine learning models handle ABSA tasks by manually generating a set of features and an aspect-category sentiment polarity classifier. Nevertheless, these traditional learning methods do not perform well in defining the complex nonlinear correlations between features and sentiment polarity. Moreover, the quality of features is essential for obtaining satisfactory results, and inadequate features can easily cause a performance bottleneck [53].

In recent years, deep learning has emerged as an alternative to traditional machine learning models, gaining increasing use in NLP tasks with outstanding results. Recently, these methods have outperformed existing models in ABSA, as they are trained end-to-end and can automatically capture significant features [54]. Thus, in this research, in addition to employing LSTM networks, we selected the most frequently used transformer-based models for sentiment classification in scholarly work: BERT, DistilBERT, XLNet, and RoBERTa.

There exist some contributions to using user-generated content, specifically reviews, to understand customer perceptions. A frequently analysed domain is the hospitality industry, where hotel reviews offer a widely available resource for understanding customer preferences. As an illustrative example, [55] used SVM for sentiment classification in hotel reviews. The SVM model achieved an average accuracy of over 80%, outperforming other machine learning classifiers such as NB and KNN.

According to [56], aspect-level analysis is necessary to obtain a fine-grained understating of hotel guest reviews. They propose an unsupervised approach to analysing hotel reviews from a leading Chinese platform, which outperforms existing models in terms of classification performance. The authors also demonstrate that important business insights, requiring immediate action from hotel managers, can be identified. [57] presents a sentiment analysis model for Vietnamese beauty product reviews. The model's architecture includes input and hidden layers with components such as Bi-LSTM, Bi-GRU, and Conv1D. They explore both single-task and multi-task learning approaches, with the best model being BiGRU + Conv1D. This model achieved a high F1-score of 98.09% for aspect detection and 91.01% for sentiment classification using the single-task learning approach.

The summary of the related contributions is presented in Table 1. First, we observe that the studies utilize a range of methodologies, including traditional machine learning models like SVM and NB, as well as more advanced techniques such as BERT, RoBERTa, and LSTM. In particular, several studies highlight the effectiveness of transformer-based models, such as BERT and RoBERTa. This underscores the growing importance of these models in capturing complex sentiment and aspect nuances in text data, as well as the need for further testing of recent developments. Second, the contributions highlight the evident need for manually annotated datasets, which are crucial for training and evaluating sentiment analysis models. However, creating such datasets is often labor-intensive and requires domain expertise. In most cases, research relies on a few existing datasets

Table 1 Summary of sentiment analysis literature review

References	Year	Method(s)	Key findings	Relevance to our study	Dataset and language	Domain	# of Aspects	Annotation method	Best F1
[8]	2022	TF-IDF & BERT	Identifying the drivers of customer satisfaction	Automatic interpretation of customer feedback	Survey	Customer feedback in the service industry	11	Manually	72%
[9]	2022	XGBoost, DT, LR, KNN, GaussianNB	Information entropy to understand customer satisfaction in e-commerce	Advanced analytical techniques in sentiment analysis of e-commerce customer reviews	Online medical APPs reviews	Medical	5	Lexicon-based	–%
[56]	2021	Unsupervised methods	Customer preferences across various hotel attributes	Identify customers' preferences from online reviews	Chinese hotels' online reviews	Hotel industry	6	Machine and manual annotation	79.13%
[22]	2020	Conditional Random Fields (CRFs) & SVMs	Combining feature extraction sentiment classification	Sentiment analysis of online reviews	English and Chinese online reviews	Mobile and car	–	Machine and manual annotation	–
[24]	2018	Unsupervised and supervised	Usefulness of unigram features	Sentiment analysis on social media text	English tweets and SMS	Tweets about companies	–	Manually	–
[26]	2020	T1-OWA operator	Product recommendations and aspects of product reviews	E-commerce user reviews and aspect-based sentiment analysis	English hotel review	Hotel industry	Experi-ence: 8 + 6	–	–
[29]	2021	BERT Adversarial Training (BAT)	Improvement robustness and accuracy	Accuracy and robustness of sentiment analysis	SemEval 2014–2016	Electronic devices and Catering	Laptop:-, Restaurant:5	Manually	81.93%

Table 1 (continued)

References	Year	Method(s)	Key findings	Relevance to our study	Dataset and language	Domain	# of Aspects	Annotation method	Best F1
[32]	2018	Doc2Vec, SVM, LR, NB	Combining Doc2Vec with classifiers	Sentiment analysis on online customer reviews	Chinese hotel reviews	Hotel industry	–	Automatically by using user rating	81.16%
[33]	2022	BERT based CBRNN	Recurrent neural networks in sentiment analysis	Analyzing social media data or reviews with transformer-based models	English reviews (Kaggle datasets)	Airline, car, politics, movies	–	Manually	94%
[37]	2020	CNN, LSTM, BERT	Effectiveness of contextualized embeddings and LSTM's ability to capture sequential dependencies	Deep learning models in sentiment analysis of online customer reviews	Drug reviews	Medical	-	Automatically by using user rating	90%
[41]	2019	BERT	General classifier model with contextual word representations	Customer satisfaction in e-commerce platforms with BERT for ABSA	SemEval	Hotel industry, Catering and Electronics	Laptop: -, Restaurant: 5	Manually	85.0%
[45]	2004	FBS (Feature-Based Summarization)	Feature-based summaries of customer reviews, opinion mining, and feature identification	ABSA of customer reviews of e-commerce platforms	Customer reviews of five electronics products	Electronic Devices	–	Manually	–

Table 1 (continued)

References	Year	Method(s)	Key findings	Relevance to our study	Dataset and language	Domain	# of Aspects	Annotation method	Best F1
[48]	2019	Joint Aspect-Based Sentiment Topic (JABST) and MaxEnt-JABST	Benefits of combining maximum entropy classification with topic models	ABSA of customer reviews of e-commerce platforms, extraction of fine-grained aspects	Online reviews	Catering and Electronics	–	Automatically	–
[49]	2014	SVM	Sentiment lexicons and word-aspect association lexicons	Aspect detection and ABSA of online reviews	Restaurant and Laptop Reviews Corpus	Catering and Electronics	–	Lexicon based and manually	–
[51]	2017	Recurrent Attention on Memory (RAM)	Leveraging multiple attentions and recurrent mechanisms	ABSA of online reviews and social media data	SemEval2014, Tweets and Chinese News comments	Catering, Electronics, Politics	–	Manually	73.85%
[52]	2021	BERT, RF, fuzzy logic	Ensemble model in sentiment classification	BERT for ABSA of online reviews	English Online hotel reviews from TripAdvisor	Hotel industry	8	Automatically by using user rating	84%
[53]	2021	RACSA (RoBERTa-based ABSA)	Leveraging RoBERTa, document attention, 1D-CNN, cross-attention mechanism	RoBERTa for online reviews	Chinese AI Challenger 2018	Catering and Traffic	5	–	74.4%

Table 1 (continued)

References	Year	Method(s)	Key findings	Relevance to our study	Dataset and language	Domain	# of Aspects	Annotation method	Best F1
[54]	2019	CapsNet, CapsNet-BERT	Capsule networks, and BERT, excels in complex multi-aspect multi-sentiment analysis	BERT for ABSA	AspectMulti-Sentiment (MAMS) dataset	Catering and Electronics	8	Manually	–
[55]	2020	NB, SVM	Analyzing customer reviews at the aspect level, detailed insights into customer satisfaction	ABSA on online reviews	Customer reviews of guest houses from travel agent websites	Hotel Industry	5	Manually	94%
[58]	2022	LSTM, RNN, ANN	Combination of online reviews and LSTM prioritizing areas for improvement	LSTM on online customer reviews	Chinese hotels' online reviews	Hotel Industry	–	Lexicon based	–
[57]	2022	Variations of BiLSTM and BiGRU combined with Conv1D layers	Effective aspect detection and sentiment classification	ABSA for e-commerce reviews	Vietnamese Reviews on E-commerce Websites	Beauty Product	7	Manually	91.01%
[59]	2021	NB, SVM, BERT, RoBERTa	RoBERTa outperformed traditional machine learning models	Sentiment analysis of e-commerce reviews	English reviews from Trustpilot	Fashion and Home product industry	–	Manually	99.2%

or employs automated annotation. Third, the review of existing research highlights the distinctiveness of our dataset, particularly in terms of the variety of aspects and domains, as such a manually annotated dataset has not been previously introduced in academic research. In conclusion, the literature review emphasizes the dynamic nature of sentiment analysis research, in particular in e-commerce, the promising results of transformer-based models, and the need for high-quality datasets with meticulously selected aspects.

3 Methodology and data

In this section, we present our NLP-based methodology for ABSA. Our primary objective is to identify the best Aspect-Based Sentiment classification Model when applied to user-generated content in the form of online reviews of e-commerce platforms. As illustrated in Fig. 1, we applied six steps to achieve our research objective. In this section, we discuss data collection, annotation, and pre-processing, and present the machine learning models used for ABSA.

Various methodologies can be used to understand and analyse user experience on e-commerce platforms. However, we can identify several advantages of our chosen methodology, NLP-based analysis of customer reviews. First, while surveys and interviews are widely used methods [60], our selected approach can analyze large volumes of data, providing a broader understanding of user experience across different products and services. Second, by analysing reviews with the proposed

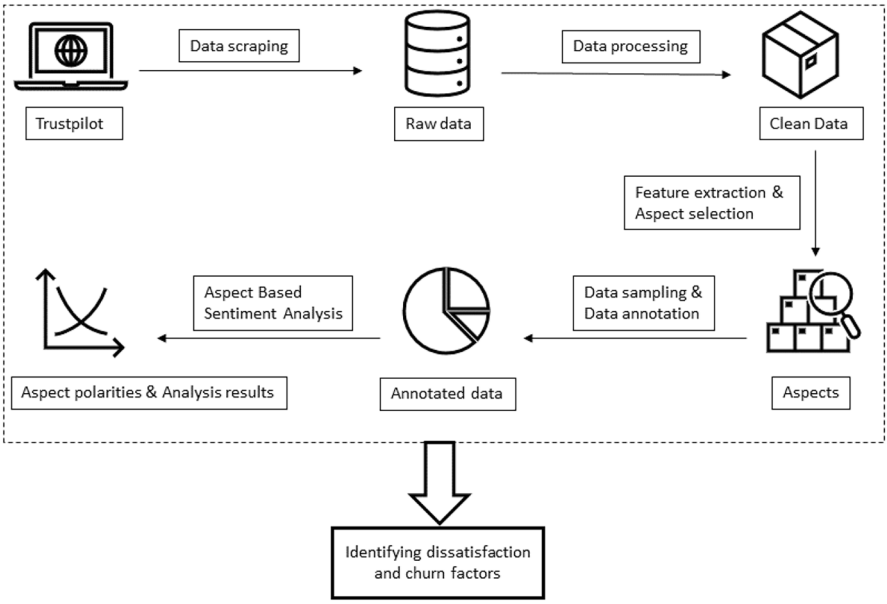


Fig. 1 The stages of the research process

method in real time, the more immediate results on user sentiments can be more insightful than delayed insights from, e.g., usability tests [61]. Additionally, the proposed machine learning models allow us to gather more unbiased insights, free from the potential biases that might affect survey responses or controlled environments in usability testing. Finally, the depth of our analysis allows for detecting subtle nuances in sentiment and aspect-based preferences that might not be captured through web analytics [62].

3.1 Data collection, preparation and aspect extraction

To create a dataset for modeling how user-generated content can help businesses to generate useful insights with ABSA, we collected 12000 English reviews from a leading online review platform Trustpilot. We collected reviews from five shops that exclusively operated online stores at the time of data collection. The stores included in the analysis are (see also Fig. 2): Zalando, Wish, SHEIN, Boozt, and Nelly. The reviews were written between 2013 and 2021 (see Fig. 3). From these, we randomly selected 3500 messages for aspect extraction and manual data annotation. We excluded messages that did not meet certain criteria, such as those not written in English or that did not discuss any of the considered aspects. The final dataset contains 2782 data points, and we will present related statistics in the following sections. Each of the selected companies operates in the e-commerce space with a focus on fashion, lifestyle, and a variety of consumer goods, serving different market segments and regions. These five e-commerce shops share several commonalities, and while each has a unique positioning and strategy, they collectively represent the dynamic nature of the e-commerce fashion industry. While Zalando and Boozt offer

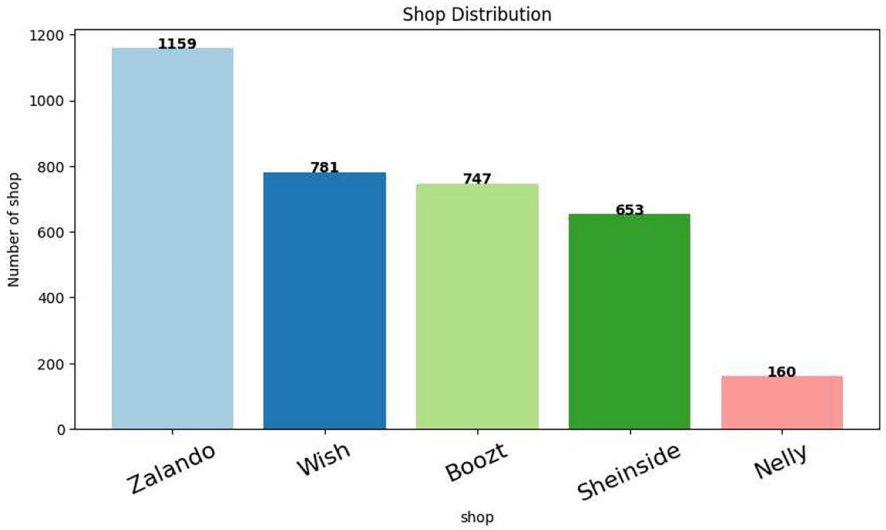


Fig. 2 The distribution of reviews by company

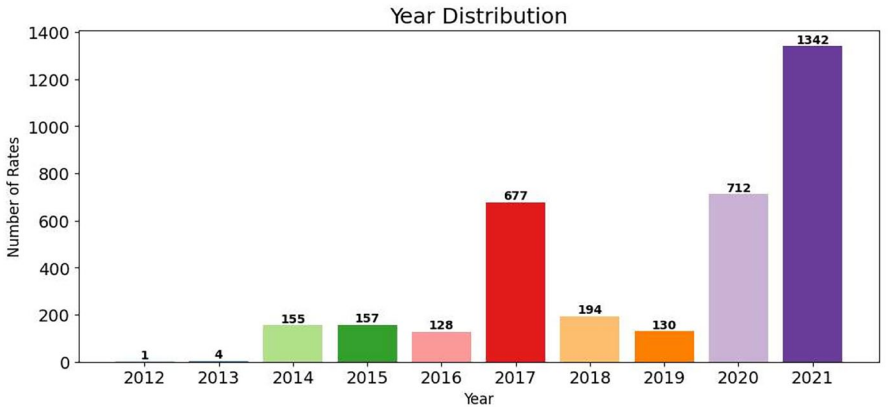


Fig. 3 The distribution of reviews by year

a wide range of clothing, shoes, and accessories, SHEIN specializes in fast fashion. These platforms serve a global market or significant regional customer base, demonstrating substantial growth in the e-commerce sector. They adapt to online shopping trends and offer a variety of brands and choices, catering to diverse fashion and lifestyle preferences. In the following, we provide some additional details about the selected platforms:

1. *Zalando*: a leading European online platform for fashion and lifestyle, founded in 2008 in Berlin. The company offers a wide range of products including clothing, footwear, and accessories (over 7000 brands). In 2022, Zalando generated revenue of 10.3 billion euros, with roughly 16,000 employees, and over 50 million active customers.[1]
2. *Wish*: founded in 2010 in San Francisco, Wish is known for its affordable goods and diverse product range, including gadgets, clothing, and home accessories. It operates as an online e-commerce platform connecting buyers and sellers worldwide (60+ countries in 5 continents). Wish maintains a substantial user base, with 24 million monthly active users and over 600,000 merchants selling on the platform.[2] In 2022, Wish reported revenue of $570 million.[3]
3. *Sheinside (SHEIN)*: established in 2012, SHEIN specializes in women's fashion but also offers men's apparel, children's clothes, accessories, shoes, bags, and other fashion items. The company is known for its fast-fashion business model with a significant presence in multiple international markets (150+ countries) and has a team of nearly 11,000 employees.[4] SHEIN, a global fast fashion e-commerce

[1] https://corporate.zalando.com/en/about-us/who-we-are/zalando-glance.
[2] https://www.wish.com/companyinfo.
[3] https://ir.wish.com/news-releases/news-release-details/wish-reports-fourth-quarter-and-fiscal-year-2022-financial.
[4] https://www.sheingroup.com/about-us/shein-at-a-glance/.

platform, has seen rapid growth, especially during the COVID pandemic, with revenues in 2022 amounting to $ 22.7 billion.[5]

4. *Boozt*: is an online fashion retailer founded in 2011, Malmö, and focused mainly on the Nordic market. It offers a wide selection of, mainly Nordic, clothing, shoes, and beauty products. Boozt consists of 1200+ employees from more than 50 nationalities.[6] The company achieved a net revenue of SEK 6.7 billion in 2022, generated mainly from Nordic brands.[7]
5. *Nelly*: a Swedish online retailer, founded in 2004 in Borås, specializing in fashion that primarily serves young adults interested in fashion and beauty.[8] The platform offers clothes, shoes, accessories, and sportswear from internationally well-known brands. As of 2022, Nelly's net revenue was approximately SEK 1,3 billion, generated by 1.1 million customers[9]

When collecting data about the five platforms, in addition to the reviews, we also gathered supplementary information such as the review header, numeric rating (range 1–5), user location, review date, and the total number of reviews previously written by the user. The reviews are written by reviewers located in 73 different countries. However, the United Kingdom (UK) with 1166 reviews, and the United States with 778 reviews, have the most total number of reviews among all the countries.

We performed seven steps to clean the text: (1) transforming the text into lowercase, (2) removing all non-English text, (3) removing all the non-alpha characters, (4) removing the HTML tags and URLs, (5) performing the contractions on the review, (6) removing the stop words, (7) performing lemmatization. After text processing, the total number of unique words is calculated as 4580.

Before performing manual data annotation, the relevant aspects of the domain needed to be specified. In addition to the perspectives identified in the literature, we aimed to extract additional aspects from the reviews. We followed three steps to extract potential aspect candidates. First, we used NLP tools to perform part-of-speech tagging on all the reviews. Afterwards, we formed bi-terms using five major combination types [63]: (Noun, Adjective), (Noun, Verb), (Noun, Adverb), (Verb, Adjective), and (Verb, Adverb). Finally, we calculated the frequency of the most frequent bi-term combinations found in the messages. We performed a manual inspection to identify the most frequent bi-terms that appeared in the reviews, as shown in Table 2.

As an example, customer service appeared directly 344 times in the reviews and also appeared indirectly in phrases such as 'great communication', 'live chat', 'never response', 'gotten answer', 'received answer', 'untruthful response' and 'untruthful

[5] https://www.businessofapps.com/data/shein-statistics/.
[6] https://careers.booztgroup.com/pages/about-us.
[7] https://www.booztgroup.com/whoweare.
[8] https://nelly.com/uk/about-nelly/.
[9] https://www.nellygroup.com/en/media/press-releases/year-end-report-2022-nelly-group-ab-nelly/.

Table 2 Most frequent bi-term combinations

Bi-term	Count
Great service	148
Quick delivery	107
Excellent service	100
Good service	91
Good quality	70
Fast delivery	52
Excellent customer-service	36
Good price	33
Great experience	33
Great price	33
Free delivery	32
Easy order	26

Table 3 Most frequent bi-term combinations for customer service aspect

Bi-term	Count
Excellent customer-service	36
Great customer-service	18
Good customer-service	15
Poor customer-service	15
Bad customer-service	12
Terrible customer-service	11
Worst customer-service	9
Horrible customer-service	8
Helpful customer-service	8
Fantastic customer-service	6
Awful customer-service	5
Friendly customer-service	5
Top customer-service	2
Responsive customer-service	2
Lie customer-service	1
Broke customer-service	1

message'. In Table 3, we present a few examples when customer service was mentioned directly in the reviews.

3.1.1 The final set of aspects

In selecting the final set of aspects, we relied on relevant contributions from the literature (such as [5, 7, 64, 65]) and the empirical analysis of the dataset described above. Ultimately, we identified the aspects presented in Table 4. We acknowledge that other aspects identified in the e-commerce literature as relevant to understanding

Table 4 The list of aspects categories and definitions

	Aspect	Description	Reference
1	Shipping	The quality, cost, and timeline of delivery are critical in shaping customer satisfaction. Prompt and cost-effective shipping enhances customer experience	[66]
2	Trust	General consumer opinion about the store, encompassing reliability and credibility play a significant role in increasing customer satisfaction	[64]
3	Item quality	The inherent quality of the products offered is a primary determinant of customer satisfaction	[64]
4	Customer service	The quality of interaction between customers and store representatives, including responsiveness and helpfulness, significantly impacts customer satisfaction	[67]
5	Pricing	Competitive pricing strategies, including discounts and campaigns, are key factors in customer decision-making	[64]
6	Product features	Detailed and accurate product descriptions, including high-quality images and size guides, aid customers in making informed purchase decisions	[65]
7	Refund process	Efficient handling of refunds, both in terms of speed and quality, directly affects customer trust and satisfaction	[68]
8	Return process	A convenient, quick, and cost-effective return process enhances the customer experience	[68]
9	App experience	The usability and user experience of the e-commerce platform, including its website and mobile app, are critical in retaining customers	[69]
10	Delivered product status	The condition of the product upon delivery, such as being intact or damaged, is a direct indicator of service quality	[65]
11	Information	The availability and accuracy of information, including clear and honest advertisements and availability of store contact information, are essential for customer satisfaction	[7]
12	Packaging	The attractiveness and quality of packaging can significantly impact customer perception of the product and brand	[5]
13	Payment	A smooth and secure financial transaction process and availability of different payment options are crucial for a positive shopping experience	[69]
14	Product availability	A wide variety and availability of products or brands cater to diverse customer needs and preferences, enhancing satisfaction	[7]

customer sentiment do not appear frequently enough in our dataset to warrant their inclusion as distinct aspects. For example, although the security of e-commerce sites is widely discussed in the literature, it appears in only a handful of reviews in our dataset. This suggests that security is not a significant concern for the population of customers who post their reviews on Trustpilot. To ensure the accuracy of our aspect selection, we randomly chose 300 messages for manual annotation. After annotation, we observed that the selected aspects correspond to the most relevant topics mentioned in the reviews. In the following, we provide a definition and justification for the final set of aspects used in the research, as summarized in Table 4.

Information, or more specifically information richness, refers to the availability (or lack) of the information that can help customers in different stages of shopping online, for example, the information presented about the products on the platform [7]. Increased amounts and better quality information have the potential to result in customers being more satisfied with their purchases. In this research, we assign this aspect anytime when it concerns any information available on the e-commerce site, including also such things as contact information (or lack thereof), (misleading) advertisements, or customer reviews.

A related aspect is the concept of general user experience when interacting with the e-commerce platform (both desktop and mobile environments). This mainly refers to website/application aesthetics and convenience and overall website design [69]. We term this aspect as App Experience, and it is used when the review mentions related positive or negative issues.

As highlighted by [64], several aspects are generally considered part of e-commerce website quality. Based on the frequency of mentions in the reviews, we selected several of these as individual aspects for our model. Product Availability [7] refers to the selection of product offerings and brands available on the platform, particularly, the availability of sufficient stock. Product features [65] refers to the variety of product offerings on the platform, including different variations of products with varying features, e.g., different sizes and colors of clothing. Pricing [64] refers to the perception of price offerings, including the availability of discounts and campaigns, as well as the general price level, typically in comparison to offline stores. Payment refers to the step in the purchase process when the financial transaction is completed. While most models consider the purchase process as a whole [64, 69], we included Payment as a separate aspect due to the frequency of reviews specifically mentioning it.

The next group of selected aspects focuses on components of the post-purchase process. Delivered Product Status, termed Delivery condition by [65], refers to the condition in which the products are delivered. Additionally to comments about broken or dirty items, this aspect also applies when some (or all) items from the package are missing, or when the package is not delivered at all. Packaging, closely related to the previous aspect, captures the perception of the attractiveness, material, and quality of the packaging in which the product arrives. The most general aspect in the post-purchase stage is Shipping [66], which is measured by how customers perceive the cost and timeliness of the delivery, whether the correct products were delivered, and how easy it was to acquire information about the package. Preliminary analysis shows that this is the most frequently mentioned issue in the reviews.

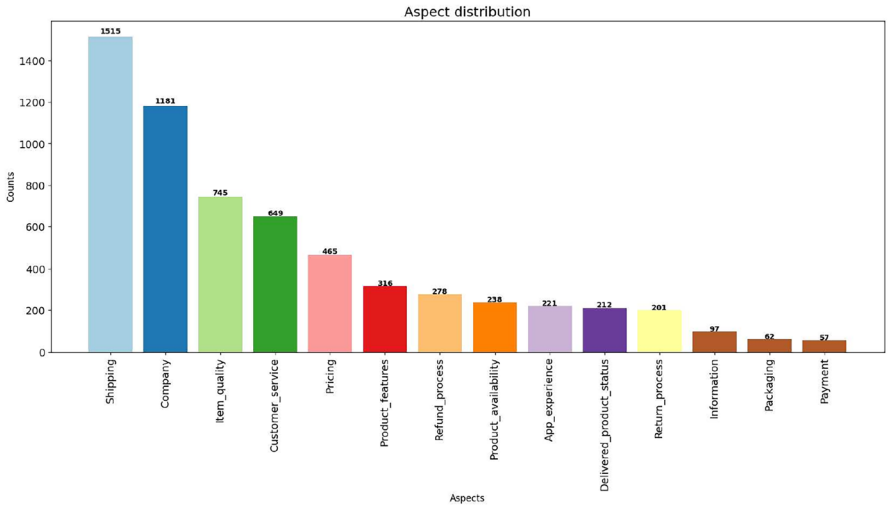


Fig. 4 The distribution of aspects

After the products are delivered, the customer evaluates the received product by assessing Item quality [64] with respect to the expectations based on the information provided when the product was ordered.

Two closely related aspects are Refund and Return [68]. Return is an essential component of e-commerce and one of the most significant differences compared to traditional commerce, as returns tend to be of much more frequent, and thus have a large impact on the overall customer experience. Reviews are assigned the Return aspect when they concern the speed, convenience, and cost of the return process. Refund, specifically refund speed and the effective handling of refund-related problems, has also been found to be important in shaping customer experience and determining customer satisfaction.

Finally, we included an aspect we termed Trust, which incorporates any issues related to the potential future use of the e-commerce platform and the customers' general opinion of the e-commerce platform, such as recommendations [64]. This includes customers' intention to purchase from the platform again and their intention to recommend the service to other potential customers (word of mouth).

Figure 4 depicts the frequency of the aspects in the dataset. As we can observe, the most frequently discussed aspect is Shipping, with 1515 mentions, accounting for approx. 50% of all reviews. This is by far the highest number among all the aspects.

3.2 Data annotation

After the set of aspects is determined, in order to perform aspect-based sentiment analysis, a polarity value needs to be appointed to each aspect-review pair. In the following, we describe how we have annotated the reviews for the ABSA problem by assigning sentiment polarities.

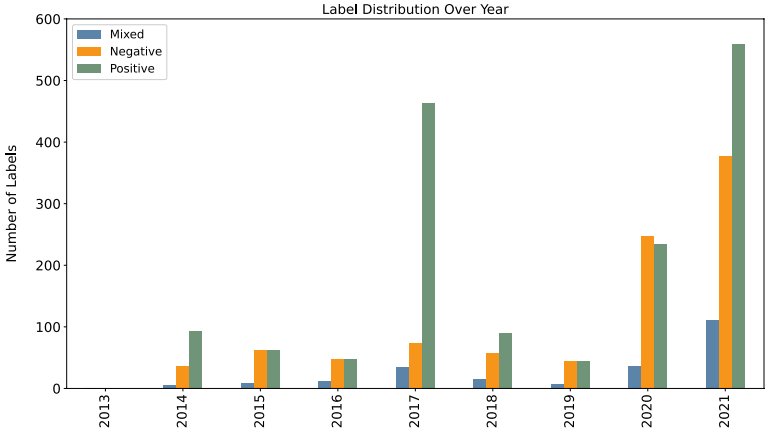


Fig. 5 The distribution of labels by year

According to the reviews selected for the study, we labeled them for the 14 aspects and the overall sentiment based on the emotions identified in the messages. In the annotation process, we used two possible sentiment labels for each aspect: positive and negative. Two annotators (two authors of the article) individually assigned sentiment values to each aspect. After the individual annotations were completed, any disagreements were marked and discussed to reach a final agreement between the annotators. Figure 5 shows the distribution of sentiment labels for each year considered in the data.

Additionally, we labeled each entire review based on the emotions identified, assigning an overall sentiment to each review as positive, negative, or mixed. We chose to perform the manual annotation for the entire messages, in addition to individual aspects, to gain a better understanding of the user reviews and the content of the messages with respect to the target companies. Some example messages with their assigned aspects and sentiment values are shown in Table 5. As illustrated in Fig. 6, there are many reviews where a disagreement can be observed between the rating (specified by the reviewer) and the actual sentiment of the review (as determined through annotation). This implies that ratings are not suitable for building sentiment classification models (as illustrated, for example, in [59]), and that more accurate measurements for model training can be achieved through manual annotation.

After the final sentiment values were determined, we discarded all reviews of unacceptable quality (e.g., those that did not express any sentiments toward any of the aspects). In other words, we continued the analysis with reviews that discussed at least one of the aspects. The final dataset contains 2782 data points, consisting of 3850 positive, and 2387 negative aspect sentiment values. As mentioned previously, the dataset includes five e-commerce platforms, and we will provide a detailed discussion on the performance of platforms with respect to the aspects. A summary of the annotations is presented in Fig. 7. As we can observe, there are significant differences in the distribution of sentiment labels across platforms. Some shops, such

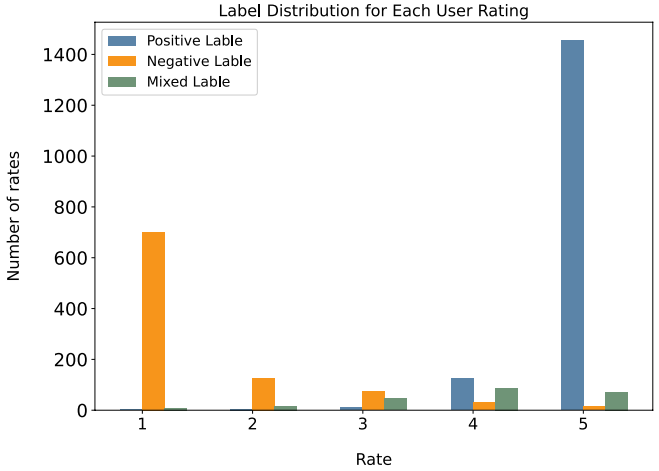


Fig. 6 Disagreements between manual annotation and user rating

as Boozt, have mainly positive reviews, while others, such as Wish, show a more mixed perception. This can also be confirmed by looking at the original rating values assigned by the customers as part of the review process, as depicted in Fig. 8.

3.3 Deep learning models

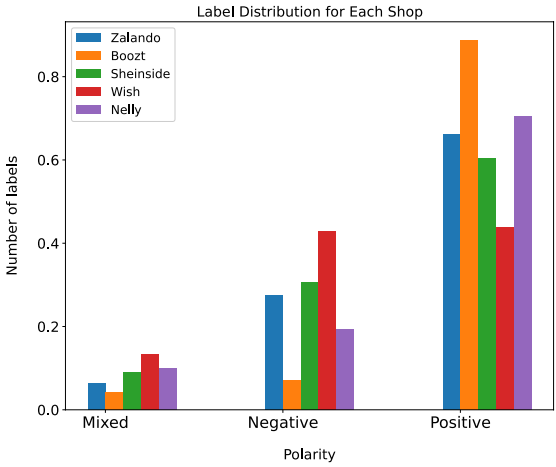
In this section, we provide details on the sentiment classification models tested in the analysis. LSTM [42], a special kind of Recurrent Neural Network (RNN), was introduced by Hochreiter and Schmidhuber in 1997 to address the vanishing gradient problem common in traditional RNNs. LSTMs are specifically designed to retain information over extended periods, making them highly effective for sequence prediction problems. They achieve this through a unique architecture consisting of multiple gates - the input, output, and forget gates - that regulate the flow of information. These gates selectively allow information to enter, be retained, or be discarded, thus enabling the network to maintain long-term dependencies in the data. This characteristic of LSTMs makes them particularly suitable for applications in sentiment analysis, where the goal is to determine the emotional tone behind a series of words.

DistilBERT [70], RoBERTa [71], and ALBERT [72] are different extensions of BERT, and they have shown sufficient performance compared to BERT in various NLP tasks. These different extensions aim to improve the performance of BERT by introducing different modifications. For instance, in RoBERTa, even though a transformer is used as the main architecture similar to BERT, it is trained differently/longer compared to BERT. Furthermore, in RoBERTa, the entire sentence is used as input and the following sentence prediction objective is removed.

XLNet [73] is a generalized autoregressive method and it focuses on the pretraining step. BERT uses the Masked Language Modeling (MLM) method in the pretraining phase, which causes a pretrain-finetune discrepancy because it ignores

Table 5 Annotation examples

Aspect	Header	Review	Annotation
Trust, product availability		It has been a very good place to shop and has lots of different items to choose from	Trust#pos, Product_availability#pos, Overall#pos
Return process, shipping, App experience	Great service	Easy to order, fast delivery and easy to return	Return_process#pos, Shipping#pos, App_experience#pos, Overall#pos
Delivered product status		Ordered two masks, only got one, the letter C did not come	Delivered_product_status#neg, Overall#neg
Product availability		Fantastic selection of products	Product_availability#pos, Overall#pos
Trust, product availability, customer service	Great company	They have fast customer support service, and they offer nice items	Trust#pos, Customer_service#pos, Product_availability#pos, Overall#pos
Refund process	Stealing money!	I have still not been refunded my third item which is from Mango via Zalando!!! It is weeks now!!!! Refund now	Refund_process#neg, Overall#neg
Shipping, item quality, product features		The jersey might take almost 4 weeks I do not know why may be due to Covid. It is usually just 2 but it was a nice qualified perfect fit jersey	Shipping#neg, Product_Feature#pos, Item_quality#pos, Overall#mix
Shipping, payment, pricing, item quality		Fantastic coat for an amazing price. Fast seamless checkout and delivery! Thoroughly impressed	Shipping#pos, Payment#pos, Pricing#pos, Item_quality#pos, Overall#pos
Shipping, delivered product status, packaging	Slow shipping, Item arrived with mold on it in a wet bag		Shipping#neg, Delivered_product_status#neg, Packaging#neg, Overall#neg
Information	Advertisements not very clear		Information#neg, Overall#neg



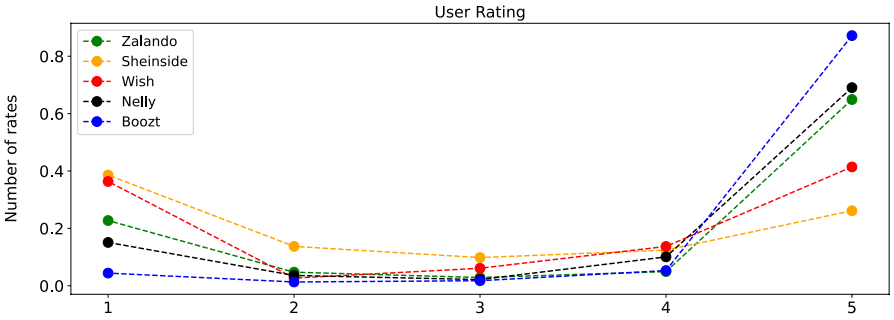


Fig. 7 The distribution of sentiment values in each shop

Fig. 8 The distribution of rating values in each shop

dependency between the masked positions. To tackle this limitation, XLNet proposed the Permutation Language Modeling objective. In addition to overcoming the constraints of BERT, by maximizing the expected probability over all permutations of the factorization order, XLNet allows learning bidirectional contexts. XLNet has been shown to outperform BERT on 20 NLP tasks [73].

DistilBERT is a transformer-based pre-trained model that was introduced to address the challenges of pretraining large models. DistilBERT is a method that pretrains a small model by applying a triple loss combining language modeling, distillation, and cosine-distance losses. Afterward, DistilBERT can be finetuned with satisfactory performances on the broad span of NLP tasks similar to the larger replications. DistilBERT demonstrates that by reducing the size of a BERT model by 40%, it is still possible to maintain 97% of its language understanding abilities and perform 60% faster [70].

3.4 Model building

To build and compare different sentiment classification models for customer reviews, in addition to LSTM, we selected four recently introduced language transformer models commonly used in sentiment analysis literature: BERT, XLNet, DistilBERT, and, RoBERTa.

Since our dataset is relatively small, we used cross-validation to obtain a more realistic estimation of our models' performance. We employed a 4-fold cross-validation process in the experiment, conducting 5 epochs for each fold. For each metric, loss, accuracy, F1 score, and AUC, we calculated the values for each fold separately. Specifically, during the training phase, we computed the training loss by averaging the loss values across all batches within an epoch and calculated the training accuracy by determining the proportion of correct predictions. We also evaluated the training AUC by applying the ROC AUC score to the predicted probabilities and the true labels. Similarly, during the validation phase, we computed the validation loss, validation accuracy, F1 score, and validation AUC using the same methodologies. After obtaining these metrics for each fold, we averaged the results to obtain a mean value for each metric per fold. Finally, we calculated the overall performance by averaging these mean values across all folds, providing a robust and comprehensive evaluation of the model's performance across different subsets of the data.

The LSTM model is built and trained for sentiment analysis using TensorFlow and Keras. First, the data is tokenized using Keras' Tokenizer, converting it into sequences of integers representing words. These sequences are then padded to a uniform length for consistent input to the neural network. Sentiment labels are converted into a Numpy array. The LSTM neural network model consists of an Embedding layer that transforms word indices into 100-dimensional vectors, an LSTM layer with 128 units for processing data sequentially and learning sequence dependencies, and a Dense layer with a sigmoid activation for binary classification. The model is compiled using the Adam optimizer and binary cross-entropy loss, and trained with K-fold cross-validation to robustly assess performance, tracking accuracy, and loss on both training and validation sets.

For transformer-based models, we utilized transformer tokenizers to convert our text input data into numeric input (AutoTokenizer for BERT, RobertaTokenizer for RoBERTa, XLNetTokenizer for XLNet, and DistilBertTokenizer for DistilBERT). For all the models, we added a dropout layer to control overfitting in addition to the baseline. When training BERT, RoBERTa, XLNET, and DistilBERT, our network model included 12 layers and 768 hidden dimensions. For all the models, the optimizer, loss function, and performance metric were specified as Adam, categorical cross-entropy, and accuracy, respectively. For the RoBERTa model, the initial learning rate was set to 3e-6, and the training batch size was defined as 8.

Figures 9 and 10 present examples of the BERT and LSTM models in Fold one of the cross-validation, with the results detailed in Tables 6 and 7.

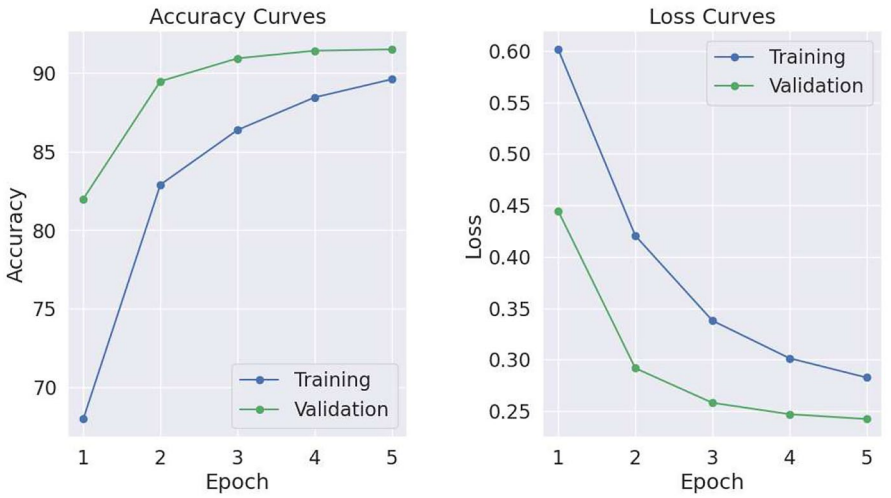


Fig. 9 The accuracy and loss in the training process for BERT

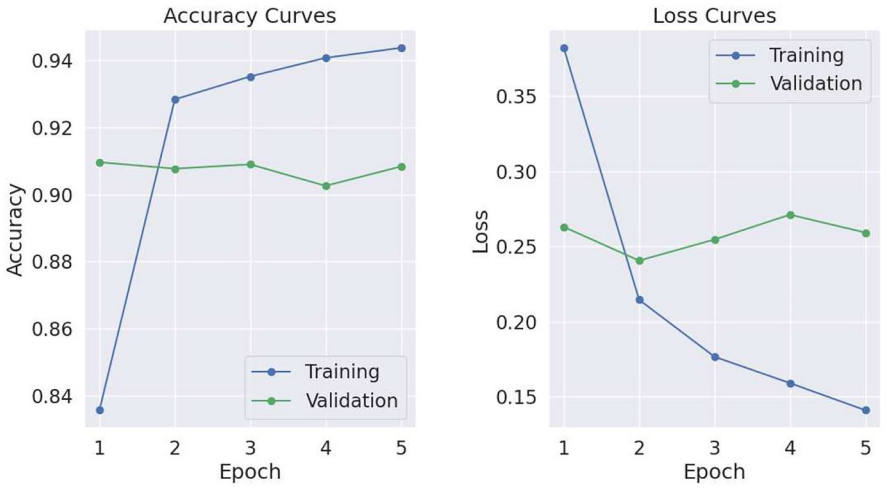


Fig. 10 The accuracy and loss in the training process for LSTM

Table 6 The performance of BERT model in each fold of cross-validation

	Accuracy (train)	Accuracy (val)	Loss (train)	Loss (val)	F1 (val)
Fold1	83.06	89.31	0.38	0.29	0.89
Fold2	83.73	89.56	0.37	0.28	0.89
Fold3	83.06	87.94	0.38	0.31	0.88
Fold4	84.47	89.15	0.35	0.29	0.89

Table 7 The performance of LSTM model in each fold of cross-validation

	Accuracy (train)	Accuracy (val)	Loss (train)	Loss (val)	F1 (val)
Fold1	91.67	90.74	0.21	0.26	0.92
Fold2	91.48	90.89	0.21	0.28	0.93
Fold3	91.62	90.55	0.21	0.27	0.92
Fold4	91.50	90.43	0.21	0.28	0.93

4 Results

In this section, we present the results of the analysis. By following the research methodology, we aim to generate high quality results in terms of sentiment classification and to utilize sentiment analysis to generate insights that would not be possible with other methods. This process allows for the systematic extraction of useful information. First, we present the results of sentiment classification and identify which model offers the best performance. Next, we illustrate the different ways in which accurate sentiment values can be used to gain a deep understanding of the data and identify the factors that have the greatest impact on customer satisfaction.

4.1 Sentiment classification

We applied three widely used evaluation metrics in the ABSA task to quantify the performance of each model: Weighted F1 score, Area Under the Curve (AUC), and Accuracy. Accuracy indicates the ratio of correctly classified instances out of all samples. The F1 score is defined as the harmonic mean of precision and recall, and is used to provide a more comprehensive evaluation of different sentiment classification models. The Receiver Operating Characteristics (ROC) curve analyzes the performance of an algorithm across all possible prior distributions and cost scenarios without committing to a specific class prior distribution or cost matrix. Its associated performance metric, AUC, is increasingly used in machine learning research, particularly in situations of class imbalance [74–76]. We built five different models as described in the previous section and the results are illustrated in Table 8. We present the accuracy and AUC for the training set and validation set, as well as the F1 value for the validation set.

The results reveal that RoBERTa outperforms the other models with an F1 score of 92.53% on the validation set. However, RoBERTa, DistilBERT, XLNet, and LSTM all performed significantly well (above 90%) compared to the BERT model (87%). A similar trend can be observed across all evaluation metrics. Furthermore, the performance evaluation clearly shows that transformer-based models performed quite satisfactorily.

After running cross-validation for all five models, we trained the best model, RoBERTa, with the optimal parameters: a learning rate of 1e−07, seven epochs, and a dropout rate of 0.5. This time, we split the dataset into training (80%), validation (12%), and test (8%) sets. The training set was used to train the RoBERTa

Table 8 The average performance of different models on validation sets in the cross-validation (on training sets in parenthesis) with the best performing model, RoBERTa, highlighted in bold

Model	Accuracy	Loss	AUC	F1
ROBERTa	**92.62 (91.34)**	**0.31 (0.29)**	**91.89 (90.45)**	**92.53**
DistilBERT	92.31 (94.64)	0.32 (0.18)	91.81 (94.22)	92.27
XLnet	92.50 (92.88)	0.37 (0.26)	91.81 (92.21)	92.41
LSTM	90.65 (91.57)	0.27 (0.21)	89.84 (93.90)	92.50
BERT	88.00 (82.62)	0.31 (0.39)	86.33 (80.84)	87.55

model, while the validation set was employed to fine-tune and validate the model's performance during the training process. Finally, the test set was reserved exclusively for error analysis. The results are presented in Fig. 11

First, we observed that the number of misclassifications on the test set was 15 out of 250 data points. When looking at the misclassifications, we noted that the model often struggled to identify mixed emotions associated with certain aspects in many of the errors. For instance, in the review "*Most of the stuff I've bought on Wish seem to be good. I've had issues with ring & clothing sizes and never received some orders*", the Product Features aspect is negative, but the model incorrectly predicted it as positive. Another common type of error was the incorrect prediction of a positive aspect in a generally negative or mixed review, such as: "*I would have given 5 stars if my delivery came the next day when it said it would but the team did well to refund me the price I paid for the express delivery*". The opinion toward the Refund Process is positive; however, the model marked it as negative. Despite these challenges, the model performed relatively well across all the shops included in the experience. The accuracy on the test set for each shop was calculated as follows: 100% for Boozt, 100% for Nelly, 88% for Sheinside, 92% for Wish, and 94% for Zalando.

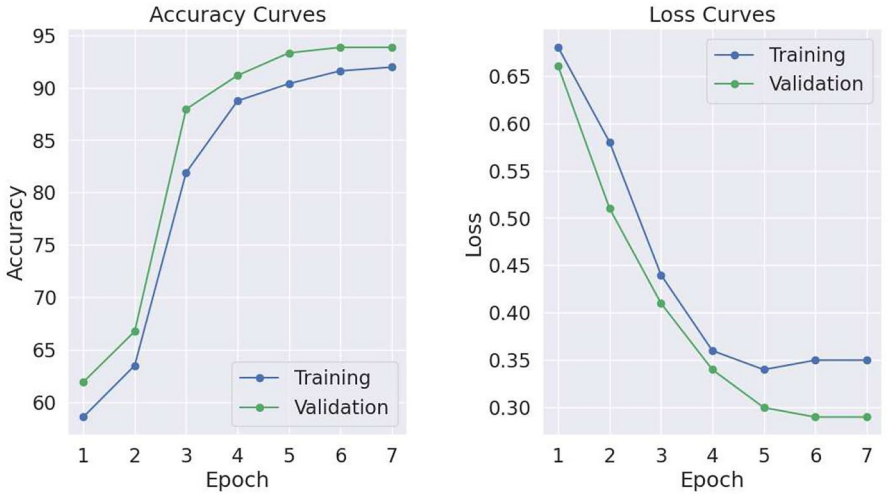


Fig. 11 The performance of the best model—RoBERTa

4.2 Understanding the most important aspects driving customer satisfaction

As the results have shown, we can achieve high performance in sentiment classification tasks. This means that if we identify a set of aspects critical to a business, we can determine whether those aspects are mentioned positively or negatively in customer reviews. The next question is how to turn this information about sentiment on different aspects into actionable insights, e.g., identifying the issues a company needs to focus on to improve customer satisfaction and, in turn, the evaluations written by the customers.

We will examine the data and sentiment of aspects from different perspectives. First, we will present some general results on the frequency and co-occurrence of aspects and sentiments to offer a high-level view of the most widely discussed components of purchasing on e-commerce platforms. We will compare the sentiment and aspect distributions for the five platforms included in the analysis to illustrate how companies can understand what differentiates them from the competition and how they can improve customer satisfaction to attract more customers. Finally, to understand more specifically which (combinations of) aspects appear more frequently in positive or negative reviews, we will present the results of an association rule analysis.

4.2.1 Understanding the distribution of aspects

First, we aimed to identify the main reasons why customers express a generally negative opinion toward an e-commerce platform. While it is also important to understand what satisfies customers, it is typically more crucial to identify the key areas a business needs to improve to enhance customer satisfaction. The most frequent combinations of the aspects that appear together in negative messages are shown in Table 9.

We observed that when sentiment toward *Shipping* is negative, most reviewers are not willing to continue doing business with the company. Another frequently mentioned aspect in the table is Trust, which reflects the feeling of the customers toward the businesses. Reviewers either state, "*I do not recommend this company and want to stop doing business with it*", or express their negative feeling by indicating, "*This is not a trusted company*". Additionally, *Customer service, Shipping* and *Refund Process* are critical aspects leading to negative evaluations. Interestingly, a

Table 9 Most frequent combinations in the negative reviews

Aspect	Count
Customer service, shipping	59
Trust, shipping	43
Trust, customer service, shipping	32
Trust, refund process, shipping	21
Refund process, shipping	12
Trust, customer service, refund process, shipping	11

Table 10 Count of aspects that appeared in the negative reviews

Aspect	Count
Shipping	509
Trust	430
Customer service	323
Refund process	183
Delivered product status	140
Item quality	108
Product features	88
Return process	72
Information	68
Pricing	48
App experience	43
Payment	31
Product availability	18
Packaging	13

negative perception of *Pricing*, whether alone or in combination with other aspects, is generally not enough for reviewers to stop using the services of these companies.

As is shown, in Table 10, *Shipping, Trust, Customer Service* and *Refund Process* are the most frequently mentioned aspects in negative messages, indicating that businesses should focus on these areas to prevent negative customer sentiment. However, the importance of other aspects should not be overlooked, as reviewers also commented on the quality of those aspects provided by the companies.

Figures 12 and 13 reveal some interesting information about the aspects. For instance, *Information* is not mentioned when the customer is satisfied; however when *Information* is poorly provided by the businesses, it can cause the overall experience to become negative. The same is true for *Delivered Product Status*: if the package is not lost, this alone is not enough for the customer to be satisfied with the overall purchase. More aspects are needed to create a pleasant shopping experience. However, if the package is lost, broken, or not as ordered, this alone is sufficient to make the customer dissatisfied. If the number of aspects that appeared in the reviews is considered alone, one might think, an aspect like *Payment* is not that important to build an unpleasant shopping experience for the customer. However, if we consider the ratio of each aspect mentioned in the messages, *Payment* ranks third.

When it comes to positive sentiments, the combination of *Item Quality* and *Shipping* appears most frequently, as shown in Table 11. We observed that *Shipping* was the most important factor for reviewers when deciding whether to continue doing business with the companies. After *Shipping, Trust, Item Quality, Pricing* and *Customer Service* were the most important aspects in the positive messages when reviewers expressed positive opinions toward the businesses, as shown in Table 12.

To differentiate the platforms included in the reviews, we performed some of the above-mentioned calculations separately for each. As Fig. 14 shows, there is a

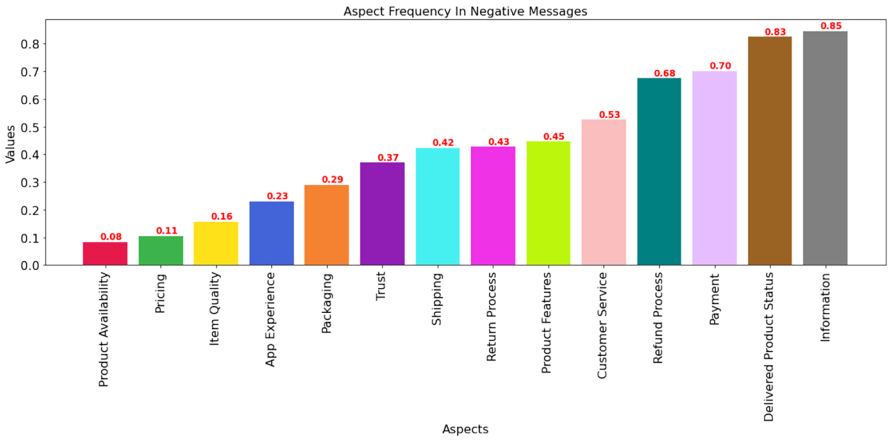


Fig. 12 The frequency of aspects in negative reviews

Table 11 Most frequent combinations in the positive reviews

Aspect	Count
Shipping, item quality	93
Trust, shipping	43
Shipping, pricing	40
Trust, shipping, item quality	33
Trust, item quality	29
Pricing, item quality	28
Shipping, product features	26
Shipping, pricing, item quality	21

Table 12 Count of aspects appeared in the positive reviews

Aspect	Count
Shipping	851
Trust	674
Item quality	566
Pricing	370
Customer service	268
Product availability	187
Product features	176
App experience	160
Return process	107
Refund process	63
Delivered product status	49
Packaging	42
Payment	19
Information	12

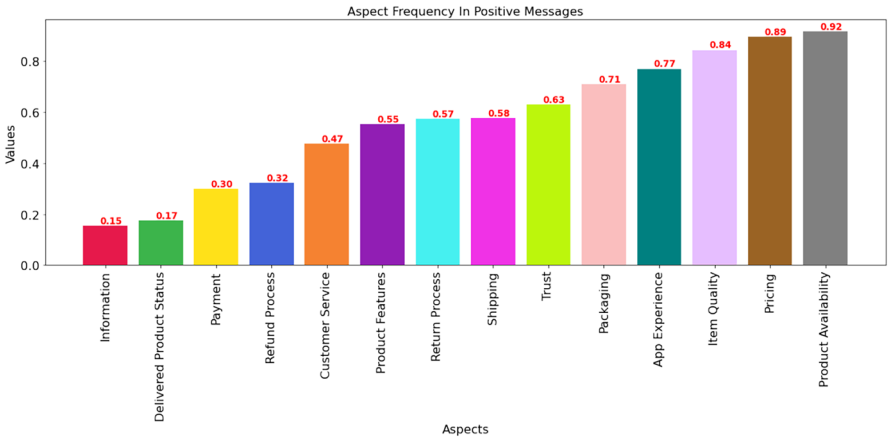


Fig. 13 The frequency of aspects in positive reviews

clear distinction in customer satisfaction across the different platforms. While one company has more than half of its reviews (60%) classified as negative, another has less than 10% negative reviews. By understanding these significant differences, poorly performing platforms can identify ways to improve their performance, in particular in terms of the aspects most important to their business.

The results regarding individual aspects and platforms are presented in Fig. 15. By looking at the differences, companies can identify the most important pain points for customers. For **Nelly**, positive sentiments dominate App Experience. This aspect is crucial for Nelly, as it can significantly enhance customer satisfaction and drive sales. Noticeable negative feedback related to Customer Service suggests that there are issues requiring attention, particularly in resolving customer problems promptly, which could improve overall satisfaction.

In the case of **Zalando**, high positive feedback related to Product Features indicates that customers appreciate the variety and features of the products offered. This is significant for Zalando, which markets a wide range of fashion items. Mixed sentiments related to Shipping suggest there is room for improvement. A more streamlined shipping process could enhance customer trust and encourage repeat purchases. For customers of **Boozt**, predominantly positive feedback on Item quality indicates that customers are satisfied, which is crucial for customer retention. However, mixed sentiments related to Shipping highlight the need for improvement to achieve a consistent and high-quality shipping service.

In the case of **Wish**, mixed sentiments suggest variability in customer trust in the company. Building and maintaining trust is crucial for customer retention, especially in a market where quality and authenticity can vary. Some negative feedback regarding Delivered Product Status highlights the need for consistent delivery quality to maintain customer interest and sales. Finally, **Sheinside** customers' negative feedback related to Customer Service and Shipping suggests issues with these processes.

Overall, we can conclude that the importance of each aspect varies depending on the type of products offered by the online shops. For instance, product quality and

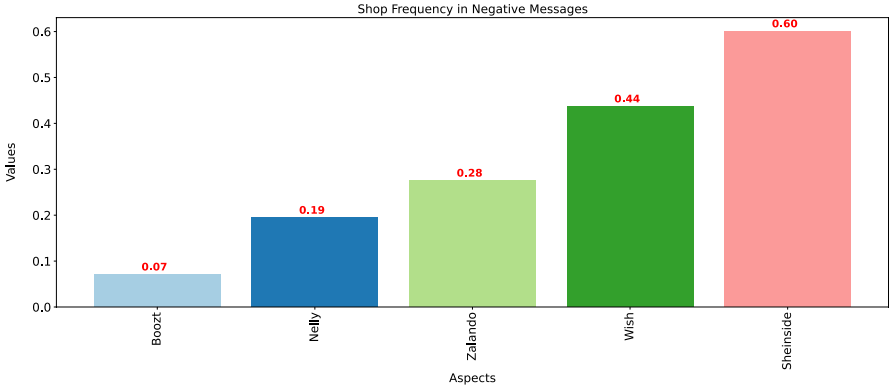


Fig. 14 The frequency of shops in negative reviews

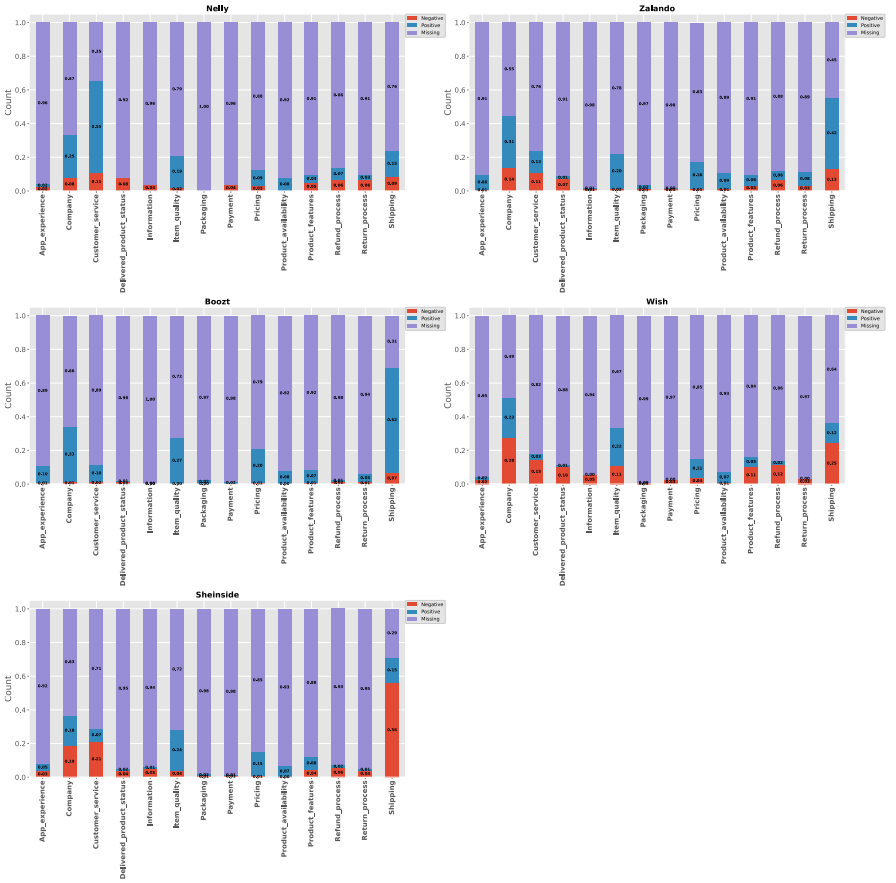


Fig. 15 The distribution of positive and negative aspects by company

features are more critical for fashion retailers like Zalando and Nelly, while trust and information may be more significant for a marketplace like Wish. Additionally, there is a heterogeneous effect of aspects across different shops. For example, while product quality consistently receives positive feedback across all shops, aspects like customer service and the refund process show more variability, indicating different levels of performance and customer expectations. By considering analysis such as the one presented in this section, companies can understand which aspects their competitors perform worse in and use this information to attract new customers.

4.2.2 Association rules analysis

We also performed an association rules analysis to discover any systematic patterns that may exist. In this context, this refers to identifying connections between the presence of certain combinations of aspects in a review and the overall sentiment. Association rules analysis [77] is the process of identifying relationships between various items based on their co-occurrence in transactions. Applied to our context, we aimed to identify combinations of aspects frequently mentioned together in reviews, and in particular those that consistently appear in either positive or negative reviews. In the terminology of association rules, we were looking for strong rules where certain aspects serve as conditions (present in the review) and positive or negative sentiment is the consequence (the corresponding review is positive or negative).

To identify the most relevant rules, it is necessary to establish threshold values for support and confidence measures, as is commonly done. Support captures the relevance of the rule; for example a rule with support 0.2 describes 20% of the data points. Confidence measures the quality of the rule; for example, if confidence is 0.9, it implies that in 90% of the reviews where the antecedents (aspects discussed) are present, the sentiment is the same (positive or negative). Our results are presented in Table 13. In this table, we listed rules that characterize at least 5% of the reviews and have a confidence of at least 80%. The first observation is that there is no combination of aspects that consistently co-occur in primarily negative reviews. This means there is no single component (or combination of components) in the e-commerce purchasing process that is consistently a source of problems. While

Table 13 Association rules with general sentiment as the outcome

	Antecedents (aspects)	Consequent (sentiment)	Support	Confidence
1	Pricing, item quality	General positive	0.063113	0.930636
2	Pricing, shipping	General positive	0.090553	0.920319
3	Product availability	General positive	0.073305	0.912195
4	Item quality, shipping	General positive	0.115641	0.891239
5	Pricing	General positive	0.145041	0.885167
6	Pricing, trust	General positive	0.069385	0.855072
7	Item quality	General positive	0.221874	0.837278

there are issues, such as shipping, that are problematic for many customers, these issues are often associated with positive general sentiment for many other customers. On the other hand, as the table shows, we can identify several combinations of aspects that frequently co-occur in positive reviews. One example is Pricing; as the confidence of rule 5 illustrates, almost 90% of the reviews where Pricing is mentioned are positive. This shows that Pricing is rarely the cause of negative customer experience on the considered e-commerce platforms. A similar, albeit slightly less consistent, pattern is observed with Item Quality.

4.3 Aspect detection dataset

To identify the position of each aspect mentioned in the text with a positive or negative sentiment, we created a manually annotated dataset [78]. The data used in this experiment is the same as what we used for manually tagging the sentiment of the aspects, but this time we also annotated the position of the aspects.

Before conducting the manual annotation, we used Python libraries like contextualSpellCheck and Caribe for spell and grammar checks to standardize the reviews. We then established rules to convert implicit aspects into explicit ones while maintaining the original text as closely as possible. After text cleaning, two annotators manually annotated the data, labeling nouns or noun phrases [79] that conveyed sentiment related to aspects, excluding neutral sentiments.

To create a dataset more suitable for aspect detection, we expanded on the previous annotation process by considering a wider range of sentence components as potential aspects and/or opinions, focusing on positive or negative sentiment within individual sentences. We used the IOB (Inside, Outside, Beginning) tagging system. The final annotation process involved specific labels to identify various aspects and marking words as the beginning or the inside of a mention, such as App_experience_B, App_experience_I, Trust_B, Trust_I, Customer_service_B, and Customer_service_I, ensuring comprehensive aspect term extraction. We used a total of 29 labels: two for each aspect and ‘O’ for words that do not belong to any aspect. In the example “*Slow shipping process. I do not recommend this company*”, the text is split into individual words, and the annotators assign corresponding labels to each word. The resulting label sequence is [‘O’, ‘Shipping_B’, ‘Shipping_I’, ‘O’, ‘O’, ‘O’, ‘O’, ‘O’, ‘Company_B’]. After completing the annotations, any disagreements were identified and resolved through discussion to reach a final consensus between the annotators.

We chose three different transformer-based models to evaluate our aspect detection dataset: BERT, RoBERTa, and BERT_Review, applying cross-validation to achieve a more accurate estimation of our models’ performance. The BERT_Review model demonstrated slightly superior performance compared to the other two models, achieving an F1 score of 84%. Considering that the study encompassed 29 distinct categories, this performance level is promising. More details about this specific modeling process can be found in [78].

4.4 Discussion

In this section, we briefly discuss the main results presented above, highlighting the novelty of our contributions. In this research, we focused on a wide range of aspects, such as Shipping, Trust, Item Quality, Customer Service, extracted from the literature and the analysed reviews. Based on this comprehensive set of aspects, we aimed to offer a nuanced understanding of the factors affecting customer satisfaction, and to identify the best machine learning model for determining the sentiment related to aspects in reviews. We conducted a comprehensive analysis of customer reviews, with a particular focus on the family of BERT-based models, in addition to the widely used RNN model, LSTM.

Our results indicate that various extensions of the basic BERT model demonstrate excellent performance in accurately determining customer sentiment across various aspects of e-commerce. While not directly comparable with existing results, as we performed our analysis on a unique dataset created for this research, the performance in terms of accuracy and F1-score is higher than what has been achieved in previous studies focusing on aspect extraction, as presented in Table 1. This finding is particularly novel, as it provides a benchmark for future research and applications in sentiment analysis within the e-commerce domain.

Comparatively, our model's performance enhances the understanding of how different neural network architectures can be optimized for the nuanced task of dissecting customer reviews into distinct sentiment-oriented aspects. This not only contributes to the academic discourse by providing empirical evidence supporting the efficacy of RoBERTa, the best-performing model, in a real-world application but also practically guides future researchers and practitioners in selecting and fine-tuning models for similar tasks.

Regarding the practical applications of our results, we can assert that the methodology enables real-time analysis of user sentiments in customer reviews, providing immediate and insightful results that are more beneficial than delayed insights from traditional methods. This could assist businesses in quickly adapting to changing customer preferences and market conditions. Complementing the machine learning models, the detailed analysis of specific aspects of customer satisfaction provides actionable insights for businesses. Understanding the relative importance of various aspects, like Shipping quality or Customer Service, can help e-commerce platforms prioritize improvements in areas most impactful for enhancing customer satisfaction and loyalty.

5 Conclusions

The use of user-generated content is becoming essential for any organization operating in the digital space. For businesses active in e-commerce, the ability to utilize and analyse such content is expected to open new opportunities to influence customers' shopping journeys. In this study, we focus on customer reviews shared by users of the platform Trustpilot, who express their opinions about five different

e-commerce platforms. We randomly selected reviews from the platform with the aim of capturing the most important antecedents of customer satisfaction for e-commerce platform users. To obtain a refined understanding, we applied aspect-based sentiment analysis with a rigorously determined set of aspects and state-of-the-art machine learning models. As our results show, one can achieve high accuracy in determining user sentiment towards various aspects of purchasing on e-commerce platforms, in particular with the RoBERTa model. We have also shown how to extract further insights from the reviews by making use of sentiment prediction and various NLP tools.

This article presents novel insights for both academia and practice. From an academic perspective, first, we present the process of manually annotating a dataset of 2782 reviews for the aspect-based sentiment analysis task to understand the roots of customer dissatisfaction in e-commerce. The set of 14 aspects used in the study was synthesized from existing academic knowledge and empirical analysis of the reviews collected from one of the most widely used review sites. To the best of our knowledge, this is the first attempt in the literature to systematically annotate a real-life dataset with a large number of aspects in this domain. This dataset can serve as a foundational resource for future research and practical applications, including in related domains, enabling more in-depth and focused research in the field.

Second, we have tested, for the first time in the literature, several of the recently introduced transformer-based models for sentiment analysis with our new dataset, establishing thresholds for future research. While there are exiting results that show how different modifications can enhance the performance of BERT and RoBERTa [53, 80, 81], we used the RoBERTa model without altering its architecture, focusing instead on fine-tuning it with our newly annotated dataset. By leveraging RoBERTa's proven capabilities in handling contextual information and applying it to the specific domain, the fine-tuned RoBERTa provides a specialized solution for aspect-based sentiment analysis in e-commerce. By conducting error analysis on the entire dataset after identifying the best model through cross-validation, we ensured that our findings are robust and provide comprehensive insights into model performance and areas for improvement. Furthermore, we have clearly shown that the original BERT model can be significantly outperformed by different variants. Overall, we demonstrate that the state-of-the-art deep learning models represent significant methodological advancement, offering a more detailed and nuanced analysis compared to traditional sentiment analysis methods. Finally, with descriptive, frequency-based analysis, and association rules, we identified which (combination of) aspects are more related to overall customer satisfaction and sentiment.

Our research approach, along with the specific findings, can provide valuable insights for practitioners. First, we demonstrated how e-commerce companies could rigorously utilize user-generated content by integrating state-of-the-art methods. While there is a lot of discussion about the usability of newly introduced deep learning methods, our paper is one of the firsts in the e-commerce domain to show that we can achieve performance acceptable for use in real-life situations. By integrating our findings into their customer feedback systems, companies can gain nuanced insights into consumer preferences and pain points. This can lead to more informed decision-making, targeted marketing strategies, and enhanced product development.

The potential benefits include heightened customer satisfaction, increased sales, and a robust competitive edge. Secondly, we have pinpointed key factors influencing e-commerce customer satisfaction and analyzed their connection to overall sentiment. These findings are invaluable for businesses aiming to pinpoint areas for improvement. Our research paves the way for more tailored shopping experiences by allowing businesses to understand and respond to individual customer sentiments more effectively. Additionally, a refined feedback loop, informed by the advanced sentiment analysis techniques, can enable businesses to swiftly adapt to customer needs and emerging market trends.

We must also note that this study has several limitations. While the dataset was randomly selected from one of the largest review sites, we cannot claim that the results can be generalized to other sets of customer reviews. In particular, we only considered five of the largest e-commerce platforms, but there are many others that could be included in a similar analysis. Furthermore, although we aimed to consider all possible aspects based on the literature and the dataset, there may be other factors that significantly impact customer satisfaction. Additionally, the distribution of extracted reviews across the five e-commerce platforms may pose a concern. While the advanced NLP models are designed to be robust across varied sample sizes, and the presented results show stable performance across platforms, we acknowledge that a more balanced distribution of reviews could provide a richer and more evenly distributed set of insights across platforms.

With regard to future research, the most important next step is to utilize larger datasets from different review sites to test and validate our results. Specifically, future research in this domain could benefit from collecting a more balanced dataset, ensuring a more uniform distribution of reviews across different e-commerce platforms. Additionally, exploring how the volume of reviews might affect sentiment analysis results could be a valuable direction for further investigation.

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Data availability statement The original data was collected from the public review site Trustpilot (www.trustpilot.com). The processed data that support the findings of this study are available at https://doi.org/10.6084/m9.figshare.24980994.v1, and the data analysis scripts are available from the corresponding author upon request.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest regarding the publication of this article. The first author received financial support for conducting this research from Jenny ja Antti Wihurin rahasto, Liikesivistysrahasto and Marcus Wallenbergin Foundation.

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